

PAPER_1674 — Light-by-Light Cross Section = $\sigma_{\text{LbL}} = \alpha^4 = \Lambda^4$ EXACT identity

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** QED **Date:** June 18, 2026 **Status:** CLOSED — EXACT formula closure (PAPER_136x broader corpus)

Observation

PAPER_1374 (PAPER_136x broader corpus) gives:

$$\sigma_{\text{LbL}} = \alpha^4 = \Lambda^4 \text{ EXACT identity}$$

Status: **EXACT formula.**

UQFF Closed Identity

$$\sigma_{\text{LbL}} = \alpha^4 = \Lambda^4 \text{ EXACT identity} \quad \text{EXACT formula}$$

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical qed measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1374
- Related: PAPER_1656-1665 (tier-24); PAPER_1646-1655 (tier-23); PAPER_1636-1645 (tier-22)
- Calculator dispatch: `calculate_paradox({"paradox": "sigma_lbl_lambda_pow_4"})`

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